



SULPHUR, NITROGEN, HALOGEN & INERT GASES — FULL MIND MAP

Chemistry · Chapter 6.C · Topic-wise · Fully Explained · Hinglish · SSC/PCS/UPSC Ready

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1. Noble Gases & Atmospheric Composition

- Noble gases Group 18 ke colourless, odourless aur mostly monoatomic gases hote hain; exam examples: He, Ne, Ar, Kr, Xe aur radioactive Rn. Actinium noble gas nahi, balki radioactive metal hai. [Q1, Q2]
- Atmosphere mein Nitrogen sabse zyada (~78–79%), Oxygen ~20.95% aur Argon ~0.93% hota hai. Given options mein Carbon dioxide ka volume-fraction sabse kam diya gaya hai. [Q3, Q4]
- Biogeochemical “gaseous cycles” ke traditional exam classification mein Nitrogen, Oxygen aur Carbon cycles aate hain; source Q5 H_2 ko dedicated gaseous cycle na hone ke roop mein key karta hai. [Q5]
- Xenon (Xe, atomic no. 54) ek rare noble gas hai. “Xenon” Greek xenos = stranger se aaya hai, isliye ise Stranger Gas bhi kaha jata hai. [Q16]

⚠ EXAM TRAP — Q3 aur Q5 mein source-style wording yaad rakho

- ▶ Q3 ka source CO_2 ko ~0.03% batata hai; modern atmospheric CO_2 concentration 0.04% se upar hai. Exam mein options/context follow karo.
- ▶ Q5 ecological-cycle terminology textbook-dependent ho sakti hai; source-answer H_2 hai, isliye isse question-specific recall ke roop mein rakho.

2. Helium — Deep-Sea Diving & Balloons

- Deep-sea divers respiration ke liye Oxygen + Helium mixture use kar sakte hain; ise Heliox kaha jata hai. Helium ki density/airway resistance low hoti hai aur high-pressure diving mein Nitrogen-related problems ko reduce karne mein help karta hai. [Q6, Q7, Q8]
- Balloons mein Helium preferred hota hai kyunki ye air se lighter aur chemically inert hai. Hydrogen bhi light hai, lekin highly flammable/explosive hone ki wajah se safety risk zyada hai. [Q23, Q24]

3. Sulphur Compound — SF_6 Geometry

- Sulphur hexafluoride, SF_6 , mein central Sulphur ke around 6 Fluorine atoms symmetrically arranged hote hain; VSEPR ke hisab se molecular geometry Octahedral hoti hai. [Q9]

4. Nitrogen Compounds — Ammonia, N_2O & Xenon Recall

- Ammonia (NH_3) polar molecule hai aur water ke saath strong hydrogen bonding kar sakta hai, isliye ye water mein bahut easily soluble hota hai. [Q10]
- Laughing Gas = Nitrous Oxide = N_2O . Iska use medical/dental procedures mein mild anaesthetic/analgesic-sedative role mein kiya jata hai. Q11–Q15 isi same recall ko different exam forms mein repeat karte hain. [Q11, Q12, Q13, Q14, Q15]
- Nitric Oxide (NO) ko Laughing Gas kehna galat hai; correct Laughing Gas N_2O hai. Bordeaux mixture, Dry Ice aur Boric-acid wali statements ke beech source Q17 isi incorrect statement ko target karta hai. [Q17]

5. Nitrogen Cycle, Aircraft Tyres & Insectivorous Plants

- Nitrogen cycle mein fixation, ammonification, nitrification aur denitrification jaise steps hote hain. Source Q18 ka point hai ki cycling ke karan lithosphere–atmosphere system mein Nitrogen ka overall amount broadly conserved rehta hai. [Q18]
- Nitrogen atmosphere ki major “permanent gas” hai (~78%). Carbon dioxide aur ozone jaise gases comparatively variable category mein rakhe jate hain. [Q19]
- Aircraft tyres mein Nitrogen use ki jati hai kyunki ye combustion support nahi karti aur high-temperature/pressure conditions mein comparatively safe, dry aur stable filling gas hai. [Q20]

- Insectivorous plants Nitrogen-deficient boggy soils mein grow karte hain; Nitrogen requirement compensate karne ke liye insects ko trap/digest karte hain. Examples: Nepenthes, Drosera, Dionaea, Utricularia. [Q21, Q22]

6. Tear-Gas Questions & Trimethylamine

- Source Q25–Q26 old exam framing mein “tear gas” ke liye Chlorine/ Cl_2 ko key karta hai. Tear-producing riot-control agents ka actual chemistry context CN, CS jaise lacrimators se juda hota hai; Chlorine khud toxic choking gas hai. [Q25, Q26]
- Dead fish ki foul smell Trimethylamine (TMA) jaise volatile amines ki wajah se hoti hai; fish tissues ke decomposition mein ye characteristic fishy odour develop hota hai. [Q27]

⚠ EXAM TRAP — Tear Gas = Chlorine ko universal fact mat mano

- ▶ Q25–Q26 ke given options/source key mein Chlorine answer diya gaya hai.
- ▶ Modern riot-control “tear gas” agents commonly CS (2-chlorobenzalmalononitrile), CN etc. hote hain; Cl_2 ko standard tear-gas agent kehna chemically misleading hai.

7. Halogens — Physical State, Reactivity & Uses

- Group 17 = Halogens. Room temperature par Fluorine aur Chlorine gases, Bromine liquid, aur Iodine solid hota hai. Isliye given solid-state question ka answer Iodine aur liquid non-metal ka classic answer Bromine hai. [Q28, Q30, Q31]
- Halogen reactivity group mein downward decrease karti hai; Fluorine sabse electronegative aur sabse reactive halogen hai. [Q29]
- Source Q32 Bromine ko “analgesic” se associate karta hai, historical bromide sedative/medicine usage ke context mein. Isse modern routine analgesic fact ke roop mein generalise mat karo. [Q32]

⚠ EXAM TRAP — Q32 historical wording hai

- ▶ Bromide salts ka historical use sedatives/anticonvulsants ke roop mein hua hai.
- ▶ Aaj ke standard analgesics mein “Bromine” ko direct common analgesic kehna current clinical classification nahi hai; exam source-context yaad rakho.

8. Selenium — Photoconductive Non-metal

- Selenium ek non-metal hai lekin Gray Selenium semiconductor/photoconductor behaviour dikhata hai; light/electromagnetic radiation absorb karne par electrical conductivity badh sakti hai. Isi wajah se ye “poor conductor nahi” wale question ka exception hai. [Q33]

9. Fluorescent Tubes, Mercury Vapour & Neon Signs

- Fluorescent tubelight mein low-pressure Mercury vapour ke saath inert gas — commonly Argon — hoti hai. Electric discharge Mercury ko excite karta hai; generated UV phosphor coating ko visible light emit karne ke liye excite karta hai. [Q34]
- Source Q35 mein Mercury vapour + Argon aur Mercury vapour + Neon dono technically possible combinations diye gaye hain; standard/common filling Argon + small Mercury vapour hota hai. [Q35]
- Q36 Neon + Mercury vapour ko low-pressure filling ke roop mein accept karta hai; Q37 mein Argon gas aur Mercury vapour dono tubelight ke actual components hain, isliye source ne multiple-correct mark kiya hai. [Q36, Q37]
- Neon discharge tubes bright coloured glow dete hain, isliye optical decoration aur advertising signs — “neon signs” — mein Neon ka classic use hota hai. [Q38]

⚠ EXAM TRAP — Tubelight options mein component vs complete mixture distinguish karo

- ▶ Typical fluorescent lamp = low-pressure inert gas (usually Argon) + a small amount of Mercury vapour.
- ▶ Isliye agar options individual components hon to Argon aur Mercury vapour dono statements true ho sakte hain; exact combination options ko dhyan se padho.

✓ Source-question mapping preserved: Q1–Q38